Course Description

Title of Course: Prompt Engineering and Generative AI Systems **Course Code: CS301**
L-T Scheme: 3-0-0 **Course Credits: 3**

Prerequisite: Python Programming, Basics of Artificial Intelligence, Machine Learning, and Natural Language Processing.

Learning Outcomes:

CS301: Prompt Engineering and Generative AI Systems	
Course Outcome	**Description**
CO1	Explain prompt engineering principles for effective human-AI interaction and applications
CO2	Apply prompt engineering techniques to generate reliable outputs for AI tasks
CO3	Analyze prompting strategies to improve AI-generated responses and reasoning workflows
CO4	Develop AI applications using OpenAI APIs, LangChain, and prompt frameworks
CO5	Evaluate prompt-based AI systems for performance, security, reliability, and ethics

Course Contents:

Unit 1: Introduction to Prompt Engineering

Fundamentals of prompt engineering and its role in interacting with Large Language Models, how prompts influence outputs, and various prompt types and generative AI basics.

Unit 2: Prompt Design Techniques

Zero-shot, few-shot, and chain-of-thought prompting, emphasizing clarity, structure, and context for reliable outputs.

Unit 3: Advanced Prompting Strategies

Prompt chaining, role-based prompting, system prompts, and multi-step reasoning workflows for complex AI tasks.

Unit 4: Tools, APIs, and Frameworks

Implementation using OpenAI API, LangChain, and integration with external systems to build end-to-end applications.

Unit 5: Applications of Prompt Engineering

Real-world applications, such as chatbots, content generation, and automation tools, focusing on scalable solutions.

Unit 6: Ethics, Security and Optimization

Prompt injection, data leakage, ethical concerns, and optimization for cost and performance.

Text Book:

1. Prompt Engineering for LLMs: The Art and Science of Building Large Language Model-Based Applications by John Berryman and Albert Ziegler. O'Reilly Media, 2024.
2. AI Engineering: Building Applications with Foundation Models by Chip Huyen. O'Reilly Media, 2025.

References:

1. Hands-On Large Language Models: Language Understanding and Generation by Jay Alammar and Maarten Grootendorst. O'Reilly Media, 2025.
2. Speech and Language Processing by Daniel Jurafsky and James H. Martin.
3. Prompt Engineering for Generative AI: Future-Proof Inputs for Reliable AI Outputs by James Phoenix and Mike Taylor. O'Reilly Media, 2024.
4. OpenAI Prompt Engineering Guide (Official Documentation).
5. LangChain Documentation (Official Documentation).
6. Research papers on Large Language Models, Prompt Engineering, and Generative AI Systems from leading AI conferences and journals.
7. NPTEL/SWAYAM Plus Course: Prompt Engineering for Generative AI. Available at: https://swayam-plus.swayam2.ac.in/courses/course-details?id=LEPL-PROMPT0824-01. Accessed on 17 July 2026.
8. NPTEL/SWAYAM Plus Course: Prompt Engineering. Available at: https://swayam-plus.swayam2.ac.in/courses/course-details?id=P_IITMPR_34. Accessed on 17 July 2026.

Evaluation Scheme:

Evaluations	**Marks**	**Remarks**
T1	15 Marks (1 Hour)	UNIT 1 & UNIT 2
T2	25 Marks (1.5 Hours)	UNIT 3 and around 30% from coverage of Test-1
T3	35 Marks (2 Hours)	UNIT 4 - 6 and around 30% from coverage of Test-2
Assignments	10 Marks	3 or 4 Assignments to be given
Quiz	5 Marks	2 or 3 quizzes
Tutorials	5 Marks	
Attendance	5 Marks	
Total	**100 Marks**	

sssCourse Conduction Plan:

Topics	Lecture Hours
UNIT I	
Introduction to Prompt Engineering, Fundamentals of Prompt Engineering and Generative AI	**2**
Large Language Models (LLMs) and Prompt-Response Mechanism	**2**
Prompt Types and Prompt Design Principles	**2**
Role of Prompt Engineering in Generative AI Systems	**1**
Hands-on Exercises and Case Studies	**1**
UNIT II	
Zero-shot Prompting	**1**
Few-shot Prompting	**2**
Chain-of-Thought Prompting	**2**
Prompt Structuring: Clarity, Context, and Constraints	**1**
Prompt Evaluation and Best Practices	**2**
UNIT III	
Prompt Chaining and Workflow Design	**2**
Role-based Prompting, System Prompting, and Persona Prompting	**2**
Multi-Step Reasoning Workflows	**2**
Complex AI Task Design and Hands-on Exercises	**2**
UNIT IV	
OpenAI API: Authentication, Text Generation, and Embeddings	**2**
LangChain: Prompt Templates, Chains, Agents, and Memory	**3**
Integration with External Systems and REST APIs	**2**
End-to-End AI Application Development, Deployment, and Case Studies	**3**
UNIT V	
AI Chatbots and Conversational AI Applications	**2**
Content Generation and Automation Tools	**2**
Scalable AI Solutions and Industrial Case Studies	**2**
UNIT VI	
Prompt Injection and Data Leakage	**2**
Ethical AI, Responsible AI, and Security Challenges	**1**
Cost and Performance Optimization	**1**
Course Review and Emerging Trends	**1**